ASSESSMENT TOOL -2

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1. Concept Map: Security Attacks, Threats, Risks, and Security Goals

SECURITY ATTACKS

Interception Modification Interruption

Confidentiality Integrity Availability

Threat Threat Threat

Data Leakage Data Tampering Service Unavailable

Financial Loss Wrong Decisions Business Downtime

SECURITY GOALS

Confidentiality Integrity Availability

a. Attack → Threat → Risk

Attack	Threat	Risk	Example
Interception	Unauthorized access	Data theft	Hacker steals customer passwords
Modification	Data alteration	Incorrect records	Bank balance changed illegally
Interruption	Service disruption	System downtime	DDoS attack on a website

b. Security Goal → Mitigation

Security Goal	Mitigation	Example
Confidentiality	Encryption	HTTPS protects online banking
Integrity	Digital Signature/Hash	Software update verification
Availability	Backup & Firewalls	Cloud servers during DDoS attacks

2. Concept Map: Substitution Techniques

SUBSTITUTION TECHNIQUES

Caesar	Monoalphabetic	Vigenère	Hill Cipher
Single Shift	Fixed Key Mapping	Multiple Keys	Matrix Key (Linear Algebra)
Simple	Moderate	Complex	Most Complex
Weak	Medium	Strong	Very Strong

Comparison

Technique	Key Usage	Complexity	Security Strength
Caesar Cipher	Single shift	Low	Very Low
Monoalphabetic Cipher	Fixed substitution	Medium	Medium
Vigenère Cipher	Repeating keyword	High	High
Hill Cipher	Matrix key	Very High	Very High

Analysis

Most Secure: Hill Cipher

Reason:

- Uses matrix multiplication.
- Encrypts multiple letters together.
- Difficult to break without the key matrix.

3. Concept Map: Classical Encryption Schemes

CLASSICAL ENCRYPTION

Substitution | Transposition
Replace Letters | Rearrange Letters
Characters Change | Positions Change
Provides Secrecy | Preserves Frequency

Similarities

- Both are classical encryption methods.
- Both require encryption and decryption keys.
- Both aim to protect plaintext.

Differences

Feature	Substitution	Transposition
Method	Replace symbols	Rearrange symbols
Data Structure	Characters change	Character order changes
Security Level	Moderate	Moderate

i. Data Structure & Security

- **Substitution:** Changes character values.
- **Transposition:** Changes character positions.

ii. More Vulnerable

Substitution Cipher

Reason:

- Vulnerable to frequency analysis.
- Letter frequencies remain predictable.

Example:

- Caesar cipher can be cracked by trying only 26 shifts.

4. Concept Map: Steganography

STEGANOGRAPHY

Data Hiding | Watermarking | Survivability

Hide Secret Information | Ownership Protection | Resist Removal & Attacks

Secure Communication | Copyright Verification | Robust Security

Relationships

- Data Hiding conceals secret information.
- Watermarking protects ownership.
- Survivability ensures hidden data remains intact.

Contribution to Secure Communication

Concept	Contribution
Data Hiding	Conceals confidential messages
Watermarking	Protects copyright and authenticity
Survivability	Ensures data survives compression and attacks

Real-world Applications

Concept	Application
Data Hiding	Secret military communication
Watermarking	Copyright protection of digital images
Survivability	Medical image authentication

5. Concept Map: Number Theory in Cryptography

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          NUMBER THEORY
               |
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 |        |        |        |        |
Extended  Modular  Fermat's  Euler's   Euler's
Euclid   Inverse  Little    Phi      Theorem
Algorithm         Theorem   Function
 |        |        |        |        |
Find GCD  Find     Fast     Counts    Generalizes
& Inverse Inverse  Modular  Coprimes   Fermat
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Power
│
└──────────────────┐
       │
   Public Key Crypto
```

a. Mathematical Connections

- Extended Euclid Algorithm → finds Modular Inverse.
- Euler's Phi Function → used in Euler's Theorem.
- Fermat's Little Theorem → special case of Euler's Theorem.
- Modular Inverse → required for RSA decryption.

b. Role in Encryption/Decryption

Concept	Role
Extended Euclid Algorithm	Computes private key
Modular Inverse	RSA key generation
Fermat's Little Theorem	Modular exponentiation
Euler's Phi Function	Calculates RSA keys
Euler's Theorem	RSA encryption/decryption

c. Most Critical Concept

Euler's Phi Function ($\varphi(n)$)

Justification:

- Determines the RSA public and private keys.
- Used to compute the modular inverse.
- Essential for generating secure encryption and decryption keys.
- Without $\varphi(n)$, RSA key generation is not possible.